

Vu Van Tung

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Summary

AI Engineer at UDATA JOINT STOCK COMPANY with hands-on experience shipping production web apps (React, Next.js) and operating an Industrial IoT platform for solar power plants. Background in AI and Computer Vision from university projects (YOLO, pose estimation, temporal models, NLP). Focused on building AI products that run reliably in production.

Experience

UDATA JOINT STOCK COMPANY **May 2026 – Present**

AI Engineer *Ha Noi, Vietnam*

Udata Marketing Website & Recruitment Backend — udata.ai — Next.js 16, Tailwind CSS v4, Sanity CMS, FastAPI, PostgreSQL, S3

- Migrated the company website from React SPA to Next.js App Router (SSR); added sitemap, robots, canonical tags, JSON-LD schema and SEO redirects from old routes.
- Integrated Sanity CMS for blog, use cases and "Book a Demo" leads so Sales/Marketing manage content without developers; bilingual VI/EN support.
- Built a recruitment portal: candidate application modal, online assessment, and HR admin dashboard (view CVs/answers, send interview invitations).
- Backend (FastAPI, SQLAlchemy, Alembic): secured the recruitment API and removed committed secrets; CV storage on S3 with local fallback; delete-application endpoint with file cleanup; HR notifications via email and Telegram; Docker Compose + Jenkins deployment.
- Built product pages and pricing for UleadGen (AI agent product) and embedded the AI chatbot widget; optimized media (WebP, posters, lazy video via IntersectionObserver) for weak mobile devices.

Udata Careers Site — career.udata.ai — React 19, Vite, Tailwind CSS v4, Google Apps Script, Docker, Jenkins, Helm

- Built the careers site from scratch (main contributor): job listings, JD preview modal, shareable job links, visitor tracking.
- Designed the application flow: PDF CV uploaded to Google Drive, recorded to Google Sheets and emailed to HR via Apps Script, with HR-email failures isolated so applications are never lost.
- Containerized with multi-stage Docker + Nginx; fixed Jenkins/Helm deployment (stuck releases, Alpine build); added GitHub Actions lint + build checks on PRs.

Nexus-E — Industrial IoT Platform for Solar Plants — uboard.udata.ai — Node-RED, MQTT, Modbus RTU/TCP, IEC 60870-5-104, ThingsBoard, Raspberry Pi, Tailscale

- Onboarded, audited and standardized 25+ solar plant sites end-to-end (edge gateway → cloud dashboards): fixed inactive inverters/meters, device relations and telemetry rollups.
- Wrote edge device templates for Sungrow Logger1000 and Solis S6 inverters (Modbus RTU over RS485 with retry handling for shared-bus CRC errors), emitting the standard telemetry contract.
- Set up IEC-104 connections for EVN grid-operator acceptance via RUT200 IPsec routing; diagnosed field issues (Modbus timeouts, offline hardware, buffer freezes); wrote the customer dashboard guide.

UGate Mini / ULead — AI Customer Support Agent (RAG) — uleadgen.net — Next.js, FastAPI, PostgreSQL

- Contributed fixes to landing-page trial lead persistence, chat widget, call notes and notifications.

AI Projects

Multi-Person Fall Detection System (Bachelor Thesis) **Mar 2026 – Sep 2026**

Individual Project — github.com/MaricoJimmy/Fall_detection

- Pipeline: video → YOLOv8-Pose → BoT-SORT tracking → 2-second pose sequence per person → TCN classifier → per-person alert; handles several people in the same frame.
- Engineered 92 features per timestep (keypoints, velocities, bounding-box shape) and a 4-block causal dilated TCN covering the full 60-step window.
- Processed 6,988 videos; on 1,034 test videos: **accuracy 93.2%, F1 92.4%, ROC-AUC 97.8%**, with a leak-free 70/15/15 split and threshold chosen on validation only.
- **Technologies:** Python, PyTorch, YOLOv8-Pose, BoT-SORT, OpenCV, NumPy

Smart CV Matching using AI & NLP **Dec 2025 – Jan 2026**

Group Project — NLP Engineer (Feature Engineering & Embedding)

- Built a resume–job description matching pipeline; extracted skills/keywords with TF-IDF and feature engineering; ranked candidates by similarity score.
- **Technologies:** Python, Scikit-learn, TF-IDF, Pandas, NumPy

Street-Sign Recognition System **Nov 2025 – Dec 2025**

Group Project — Small-object detection

- Trained YOLOv8/YOLOv11 on the Zalo Traffic Sign dataset; applied SAHI sliced inference to detect small signs in high-resolution images; dataset preprocessing and augmentation.
- **Technologies:** Python, YOLOv8, YOLOv11, SAHI, OpenCV

Hand Tracking & Gesture-Based Object Guessing **2026**

Individual Projects

- Real-time finger counting and mouse control with MediaPipe Hand Landmarker; a gesture-triggered app that sends camera frames to Google Gemini to recognize the object being held.
- **Technologies:** Python, MediaPipe, OpenCV, Gemini API

Technical Skills

Languages: Python, JavaScript, TypeScript, HTML/CSS, C, Java

AI / ML: PyTorch, TensorFlow, Scikit-learn, YOLOv8/v11, YOLOv8-Pose, SAHI, MediaPipe, OpenCV, NLP (TF-IDF), Gemini API

Frontend: React, Next.js (App Router, SSR), Vite, Tailwind CSS, Framer Motion, Sanity CMS

IoT: Node-RED, MQTT, Modbus RTU/TCP, IEC 60870-5-104, ThingsBoard, Raspberry Pi

Backend: FastAPI, SQLAlchemy, Alembic, PostgreSQL, AWS S3, Google Apps Script

DevOps & Tools: Git/GitHub, Docker, Nginx, Jenkins, Helm, GitHub Actions, Tailscale, Google Colab

Education

University of Science and Technology of Hanoi (USTH) **2022 – 2026**

Bachelor of Information Technology *Ha Noi, Vietnam*

- **Achievements:** Final round of the Student AI Technology Competition 2025.
- **Activities:** USTH Job Fair; Technology Entrepreneurship: Thinking & Experimentation course.

Languages

Vietnamese: Native | **English:** Basic (reading technical documents)